# Industrial Vision

## Lecture 4 - From Edges to Features

Interest points and the Harris structure matrix

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# Where we are

- In the last lecture we converted intensity gradients into **edges** using Canny, and grouped edge pixels into **lines** using the Hough transform. Edges are the first genuine structure extracted from raw pixels.
- An edge pixel, however, **cannot be matched between two images**, because every pixel along the edge looks the same. This is the *aperture problem*. To align two photographs of a scene we require points that are unambiguous in **every** direction.
- Such points are called **features**, or interest points, and include corners and blobs.




Today we consider why features are needed, what makes a feature a good one, and the first major detector, **Harris**, as far as the 2×2 matrix that encodes a patch.

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# Outline of this lecture

1. We ask why features are needed, and list the tasks that require them, namely mosaicing, stereo, recognition and tracking.
2. We distinguish the **detector** from the **descriptor**, which perform two separate functions.
3. We state the six properties that a good feature should possess.
4. We contrast **invariance** with **covariance**, the two ways in which a feature must behave.
5. We develop the intuition for corners by comparing flat regions, edges and corners under a test of appearance change.
6. We derive the mathematics, proceeding from the SSD change function through a **Taylor** expansion to the **structure matrix** $R$.

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# Why do we need features?

**Mosaicing and panoramas** The same points are located in overlapping photographs, the alignment between them is recovered as a **homography**, and the images are stitched together. No depth information is required, since the problem is purely two-dimensional.

**Stereo and multi-view geometry** A point in 3D projects into two images. If it is matched in both, the two rays can be extended and its depth recovered by **triangulation**. Matching is thus a prerequisite for any 3D reconstruction.

**Recognition under occlusion** A telephone that is half hidden behind a shoe still shares enough local features with the unoccluded object to be recognized.

**Motion and tracking** A point is tracked across the frames of a video, which requires features that survive small motions and changes of illumination.

Each of these tasks separates into **two distinct problems**.

**detector** → **descriptor**

where are the interesting points? — how are they matched?

# The detector and the descriptor are distinct

## Detector

The detector indicates **where** the interesting points are located. A feature is something we wish to **track**, and it should therefore move together with the image.

## Descriptor

The descriptor is a vector that **summarizes the neighborhood** so that two features may be **matched**. For matching it should be independent of nuisance variation.

These two requirements pull in opposite directions.

- The **detector** should be **covariant**. If the image is rotated, the corner rotates with it, and this is precisely how the transformation is recovered.
- The **descriptor** should be **invariant**. The matching vector should not change when the lighting or the viewpoint varies slightly.

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# The properties of a good feature

**Repeatability** The feature is detected again under a slightly different viewpoint or illumination. Without this property no matching is possible.

**Distinctiveness** The feature stands out from its surroundings, so that it is easy to match and few false pairings arise.

**Locality** The feature is derived from a small neighborhood, which makes it robust to **occlusion and clutter**. A global feature, such as one obtained by PCA, fails as soon as part of the object is hidden.

**Quantity** A sufficient number of features is available. Matching is noisy, and robustness through outlier rejection depends on having many of them.

**Accuracy** The feature is precisely **localized**. Feature positions are later used as input to geometric computations, so accuracy at the sub-pixel level matters.

**Efficiency** The feature can be computed over a full image in close to real time.

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# Invariance vs covariance

An image is altered by two kinds of change, **photometric** change due to lighting and **geometric** change due to viewpoint.

**Invariance** to photometric change A change of brightness should **not** alter the feature. Otherwise a scene photographed by day could never be matched to the same scene by night.

$$\phi(\text{brighten}(f)) = \phi(f)$$

**Covariance** with geometric change If the image is rotated, the corner must **rotate with it**. This shared motion is what allows the transformation to be determined.

$$\phi(g \cdot f) = g \cdot \phi(f)$$

Here $g$ denotes a geometric transformation, such as a rotation, an affine map or a change of scale. The term covariant is not used here in the statistical sense, but in the sense of varying together with. A detector that is **covariant to rotation**, combined with a descriptor that is **invariant to rotation**, is what allows corresponding corners both to be found and to be matched.

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# The intuition behind a corner

We take a small window, **shift it slightly**, and ask how much the patch changes.

**Flat region** A shift in any direction produces **no change**.

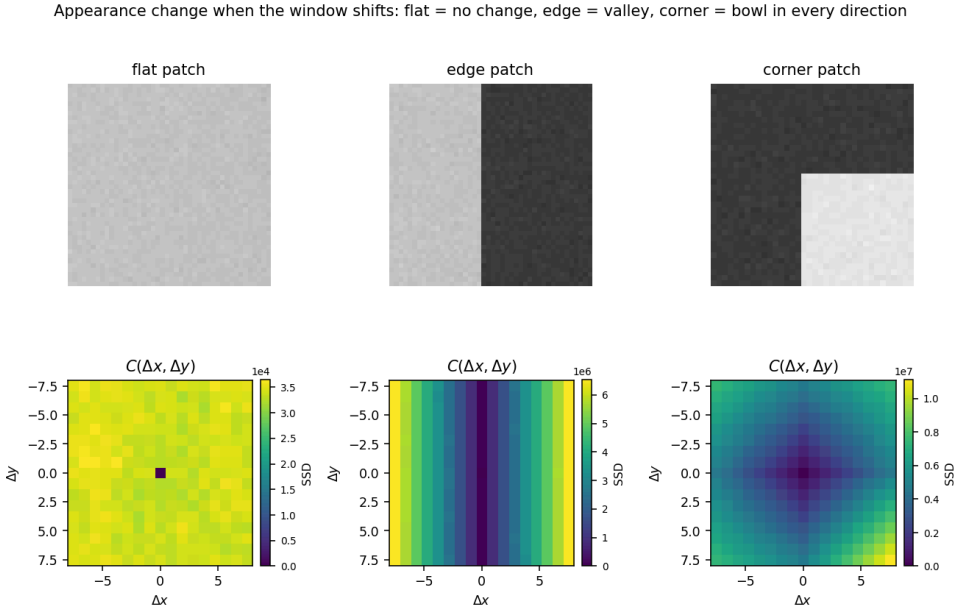
**Edge** A shift along the edge produces no change, whereas a shift across it produces a large change.

**Corner** A shift in **any** direction produces a **large change**.

A **corner** is the only patch whose appearance changes in whichever direction it is shifted. This property of changing in every direction is exactly what the Harris detector measures.

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# Measuring appearance change




The bottom row shows the change surface C(Δx, Δy), in which light color indicates a small change. The flat region is light everywhere. The edge has a dark valley, along which the window may be shifted freely. The corner is a bowl with a single minimum.

# The SSD change function

Let $(\Delta x, \Delta y)$ denote the shift. Over a window $W$ the change is measured by the **sum of squared differences**:

$$C(\Delta x, \Delta y) \;=\; \sum_{(x_i, y_i) \in W} \Big[\, I(x_i, y_i) \;-\; I(x_i + \Delta x,\; y_i + \Delta y) \,\Big]^2$$

A large value of $C$ in every direction indicates a corner. Evaluating $C$ by actually shifting the window for every candidate displacement is expensive, however, and we would prefer an expression that can be analyzed.

We use the same device as in the derivation of the Laplacian in L02, and expand the shifted intensity in a **first-order Taylor series**.

$$I(x_i + \Delta x,\; y_i + \Delta y) \;\approx\; I(x_i, y_i) \;+\; \Delta x\, I_x(x_i, y_i) \;+\; \Delta y\, I_y(x_i, y_i)$$

Here $I_x$ and $I_y$ are the image gradients, and the higher-order terms have been discarded.

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# Substitute and collect

Substituting the Taylor expansion into $C$ causes the terms $I(x_i, y_i)$ to cancel:

$$C(\Delta x, \Delta y) \;\approx\; \sum_{x_i, y_i} \Big( \Delta x\, I_x + \Delta y\, I_y \Big)^2$$

Expanding the square, in which every quantity is a scalar at each pixel, gives

$$= \sum \Big[ \Delta x^2\, I_x^2 \;+\; \Delta y^2\, I_y^2 \;+\; 2\, \Delta x\, \Delta y\, I_x I_y \Big]$$

This is a **quadratic form**, and it may be written in matrix and vector notation as

$$C(\Delta x, \Delta y) \;\approx\; \begin{bmatrix} \Delta x & \Delta y \end{bmatrix} \left( \sum_{x_i, y_i} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} \right) \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}$$

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# The structure matrix $R$

$$C(\Delta x, \Delta y) \approx \begin{bmatrix} \Delta x & \Delta y \end{bmatrix} R \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \qquad R = \sum_{x_i, y_i} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

This matrix is also called the **second-moment matrix** or the **gradient matrix**. It is constructed **only from the gradients inside the window**, and the entire distinction between a flat region, an edge and a corner is contained in the three quantities $\sum I_x^2, \sum I_y^2$ and $\sum I_x I_y$.

The gradients are computed with the small kernels of L01:

$$I_x : \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}, \quad I_y : \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}^\top$$

The Sobel kernels may also be used, since they include the smoothing factor $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix}^\top$.

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# Two Gaussians are present in $R$

**(1) Smoothed gradients.** Differentiation amplifies noise, as shown in L03, so the derivative of a Gaussian is used:

$$I_x = \frac{\partial}{\partial x}\big(G(x, y; \sigma_s) * I\big), \qquad I_y = \frac{\partial}{\partial y}\big(G(x, y; \sigma_s) * I\big)$$

**(2) A Gaussian window** $w(x_i, y_i)$ gives more weight to the center of the patch than to its rim, and so provides a soft window rather than a hard box:

$$R = \sum_{x_i, y_i} w(x_i, y_i) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, \qquad w(x_i, y_i) = e^{-\frac{(x - x_i)^2 + (y - y_i)^2}{2\sigma_w^2}}$$

Here $(x_i, y_i)$ is a pixel of the window, as before, and $(x, y)$ is the **center of the window**, so that the exponent measures the distance of the pixel from that center.

Typical values are $\sigma_w \approx 1$ for the window and $\sigma_s \approx 2$ for the gradient smoothing. The two Gaussians are implementation details, and the object of interest remains the $2 \times 2$ matrix $R$.

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# $R$ can be diagonalized

- $R$ is **symmetric**, since the two off-diagonal entries are the same sum $\sum I_x I_y$.
  - The Gaussian window does not disturb this: $w(x_i, y_i)$ is a **per-pixel scalar** inside the sum, so both off-diagonal entries remain $\sum w\, I_x I_y$. As $w > 0$, the matrix also stays positive semi-definite.
- A real symmetric matrix, $RR^\top = R^\top R$, gives real eigenvalues and orthonormal eigenvectors.

$$R = P \Lambda P^\top = P \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} P^\top, \qquad P^\top P = I$$

- The **columns of** $P$ are the eigenvectors and $\lambda_1$, $\lambda_2$ are the eigenvalues, satisfying $R\, p_i = \lambda_i\, p_i$.

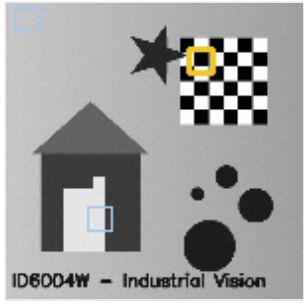
The matrix $R$ behaves like a **covariance matrix of the gradients**. Its eigenvectors point along the directions of largest and smallest intensity variation, and its eigenvalues measure the magnitude of that variation. This is precisely the information that distinguishes a flat region from an edge and from a corner, and it is interpreted in the next lecture.

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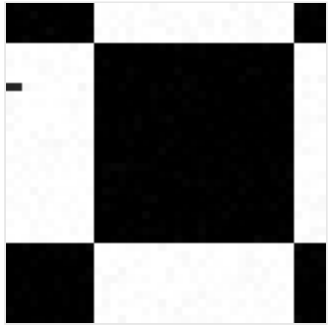
# Demo

image Synthetic scene *ideal step edges, exact corners — the textbook case*

Flat Edge Corner *windows found automatically on the selected image*




where the windows sit



patch (40×40 px)



C(Δx, Δy) — dark = small change

structure matrix eigenvalues

$\lambda_1 = 153.4M$

$\lambda_2 = 151.6M$

flat: both ≈ 0 · edge: one large · corner: both large

When a patch is selected, the demonstration constructs R from its gradients and plots the quadratic-form change surface $[\Delta x\ \Delta y]\ R\ [\Delta x\ \Delta y]^T$ together with the eigenvalues.

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# Questions

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# Summary

- A **feature** is a locally distinctive and repeatable point that can be **detected** and subsequently **described**. Features are the entry point to matching, to geometry and to 3D reconstruction.
- A good feature is **repeatable, distinctive, local, plentiful, accurate and efficient**. Detectors should be **covariant** with geometric change and descriptors **invariant** to photometric change.
- A patch is a corner if shifting it in any direction changes its appearance substantially.
- This test is formalized by the SSD function $C(\Delta x, \Delta y)$. A first-order **Taylor** expansion converts it into the quadratic form $C \approx [\Delta x\ \Delta y]\, R\, [\Delta x\ \Delta y]^\top$, in which the **structure matrix** is $R = \sum \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$ .
- Since $R$ is symmetric it admits the decomposition $R = P \Lambda P^\top$, and **everything that characterizes the patch is contained in $\lambda_1$ and $\lambda_2$**.

**In the next lecture** we interpret $R$. The eigenvalues are combined into a single corner-response score, in the forms proposed by Harris and by Shi and Tomasi. We then treat non-maximum suppression, **sub-pixel** refinement, and the invariances that the Harris detector does and does not possess.

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# References


1. Harris, Chris, and Mike Stephens. "A combined corner and edge detector." Alvey vision conference. Vol. 15. No. 50. 1988.
2. Forsyth, David A., and Jean Ponce. Industrial Vision: a modern approach. prentice hall professional technical reference, 2002. See 5.3 on corners and interest points.


Demos: `lec04_structure_matrix.py`

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